

Quants go mad in March

Every March, work in US offices grinds to a halt as employee attention and efforts are directed toward NCAA men's basketball tournament bracket pools. The tournament is a 64-team,¹ single-elimination affair. 63 games each eliminate one team, until a national champion is crowned. More important, millions of people predict the results of all 63 games in advance in pools ranging from \$1 buy-in to \$10,000, and a handful of participants to millions.

Quants go a little mad too, because NCAA pools offer lots of fascinating theoretical and empirical challenges. Since 1985, I've been running complex analyses and submitting entries to hundreds of pools. But I'm more interested in education than making a living from NCAA pools and I've been frustrated attempting to turn quant insights into a workable system for everyone—one that laid bare the mathematical principles involved as well as helped people win without demanding a lot of math or work.

This year I decided to try factor investing—basically that means applying simple dumb rules systematically rather than doing complex analyses with tailored parameters and sophisticated weightings. This has been shown to work in many fields. If you ask an expert how she makes decision, she'll give you a lot of rules, but then say her true genius comes from long experience knowing when to trust certain items and how to adjust for hard-to-quantify factors. Program the simple rules in a computer without any of the experience, complexity or adjustment and it generally does better than the expert. What the expert thinks is their important contribution to the process actually just messes up the simple, dumb rules.

Surprise! The same thing was true for me. When I took my simple, dumb NCAA tournament pool rules and applied them to a large sample of pools I had entered over the years, the rules beat my clever and sophisticated algorithms based on careful data analysis and deep theoretical considerations. And the simple, dumb rules are easy to put in a spreadsheet for anyone to use. Will they continue to do well in the future? I don't know, nobody knows. But that's why they call it "gambling."

Factor investing

Factor investing in equities goes back to the 1960s. Finance professors had just gathered overwhelming statistical evidence that professional investors could not beat the market and that expected stock returns depended only on exposure to a single factor—what the overall market did. If you knew a stock's exposure to that factor (which was named "beta") you didn't need to know anything about its business, dividends, price-earnings ratio or anything else.

But once researchers had adjusted for beta, anomalies showed up. Small stocks did slightly better than big stocks. Low price-to-book stocks did better than high price-to-book. You couldn't notice this in the raw data, only after you had adjusted for Beta did it become clear.

Of course, professional investors had always looked at these factors, but like the experts mentioned above, they insisted it required long experience and innate talent to interpret them, and that there were hundreds of factors. Professors started testing simple, dumb versions of a few factors that seemed to work well across a wide variety of markets and eras.

For example a slightly oversimplified version of a reasonable Fama-French² factor portfolio is to start

¹There are actually 68 teams, four of which are eliminated in "play-in" games that are usually ignored for bracket pools.

² Eugene Fama, the "father of efficient markets," was the empirical leader establishing the original single-factor model.

with the market portfolio then overweight by 20% stocks in the lowest 30% for price-to-book ratio and underweight by 20% stocks in the highest 30%. This is called the “value” factor but don’t do complicated estimates of value, just move the weights up or down based on price-to-book. Then move weights up 20% for the 30% smallest market cap stocks and down 20% for the 30% largest. Do the same based on prior 1-year return, operating profitability and leverage.

Fama and French actually do a somewhat more complicated version, but it’s all based on simple, mechanical rules, systematically applied, with no human judgment or excessive trading. Quant equity hedge funds do even more sophisticated versions, but it’s still based on the idea of a few underlying factors implemented with mechanical rules.

Since 1963, the stock market has turned \$1 into \$216 before fees, expenses and taxes, and nearly all professional equity investors not named Warren Buffett have done worse. The Fama French portfolio turned \$1 into \$2,279.

Professional sports gamblers reached a similar insight decades before finance professors. You don’t find simple rules like, “bet the underdog at home,” or “bet in the direction the line is moving,” that are profitable after risk and cost. Both those rules, and many others, work in a variety of sports betting situations, but not enough to overcome the vigorish or bookmaker’s commission. There’s an obvious reason for that, if they did work, the bookmakers would go out of business.

I discovered this for myself in fifth grade. I was one of those weird kids who read newspapers from the back, not because the comics were back there, but because the numbers were: sports results, stock market prices, commodity statistics and so on. I noticed that you got better payoffs at the local racetrack betting the two favorites in the exacta than you would get parlaying the win/place odds. I carefully tabulated all races for a year and discovered that the strategy of betting the favorite exacta overcame the track advantage—15% at the time—and almost exactly broke even. A statistical success, a financial failure. It broke even so precisely that I theorized some professional bettor was probably watching the money flow and grabbing any advantage in either direction. The track didn’t care, it got its 15% whoever won. The betting public was willing to lose money on average for the pleasure of hitting the rare longshot windfall.

Therefore it generally doesn’t pay to devote enormous effort to making a rule that works 51% or 52% of the time into one that works enough to overcome the bookmaker’s edge. What most professional quant sports bettors do is look for a handful, say five or so, 52% rules that combine into a 57% rules. The 52% rules are simple and dumb because the magic happens in the combination.

In finance the same idea is called “limits to arbitrage.” In theory, efficient markets should eliminate all opportunities to make market-beating profits. In practice, exploiting and correcting arbitrages is not easy, cheap or riskless. So arbitrages tend not to go away completely, but to be pushed down below the level where investors can make profits using just one. But combine five limited-arbitrage ideas in a single portfolio, and you can beat the market consistently.

In NCAA pools the untested argument is that lots of people exploit individual factors. Some people spend a lot of time estimating win probabilities for individual games. Others think about picks that may be overlooked by most other pool participants. Another type of quant analyzes the pool from a combinatoric perspective to identify over- and under-populated zones (that was my main practical focus, although I looked at the other stuff as well). Some people used game theory (this was my main

When he and co-author Ken French switched to multi-factor models in a classic 1993 paper, Fama wrote, “Beta is dead.” Fama and French have continued to publish seminal research in this area.

theoretical focus, although it seldom generated useful picks in practice).

Unlike sports betting and financial market, all these individual approaches generate net profit above costs, because there essentially are no costs, and no bookmaker to go out of business. So quants win at well above random rates. But what if combining very simple versions of a few of these approaches, in a simple way, could do better than any complex elaboration of one rule, or sophisticated combination? Let's find out!

Quality

The first factor I tried is quality, which I measure by ESPN's Basketball Power Index (BPI). I have official values from the end of the regular season since 2008, for prior years I use my own calculation that mimics the BPI. The idea of the BPI is if you subtract one team's BPI from the others, you get the expected point differential in a game. If a 10 BPI team plays a 3 BPI team, it should win by 7 points on average.

Of course, everyone knows that higher-BPI teams are more likely to win games. But the NCAA selection committee takes BPI and many other factors into account when setting seeds. In each tournament, there are four teams that share each seed from 1 to 16. The four number 1 seeds are the four best teams in the tournament in the judgment of the selection committee, the four 16 seeds are the least good teams.

For all tournaments since 2008, I split the four teams in each seed into the two with the highest BPIs and the two with the lowest BPIs. So my high-BPI group has 32 teams, 2 of each seed, as does my low-BPI group. From 1986 to 2017, teams in the high BPI group won an average of 1.35 games in the NCAA tournaments, teams in the low BPI group won an average of 0.58.³ 28 of the 32 champions came from the high-BPI group.

The direction of the result is not surprising. Most pool participants consider some sort of power rating or win probability estimate when making selections. But the effect is bigger than at least I would have expected. Within a seed teams usually have similar BPIs, you would expect the selection committee to assign teams of roughly equal quality—measured over all factors, not just BPI—in the same seed.

But I could say the same thing about quality in stocks. You would expect the market to observe operating profitability of companies, and adjust stock prices to take that into account along with many other factors. However if you buy stocks with strong operating profitability and short stocks with weak operating profitability, you make consistent profits.

Size

The size factor was one of the first factors (called “anomalies” at the time) to be documented in equity markets. Stocks with small market capitalizations outperform stocks with large market capitalizations. While it might be possible to put a franchise value on college basketball teams—successful ones can certainly generate a lot of revenue—that doesn't seem the correct translation.

Small cap stocks likely owe their return premium to being less convenient for investors. They are usually less liquid than large caps, with less information available. They are less likely to be familiar names, and typically have fewer analysts covering them. Their smaller size makes them more subject to idiosyncratic shocks.

³ There are 63 wins per tournament and 64 teams, so the average team wins 0.98 games.

Liquidity is not a factor in NCAA bracket pools, it's as easy to pick one team as another. But information, familiarity and idiosyncrasy are. So I chose to measure the equivalent of size by NCAA tournament success of the school over prior years. I gave two points for a win in round 1, four in round 2 and so on up to 64 points for winning a championship. In each round a loss gave half as many points as a win. Each year I took the previous year's value times 0.8 and added 0.2 times the current year value, and used that as a size measure for the subsequent year's tournament.

As with quality, size differs by seed. The 1 seeds are usually “big” schools, in the sense of having a lot of recent tournament success, while the 16 seeds are much more likely to be “small” schools. A 16 seed for example, with one tournament invitation in its history, a first-round loss four years ago, has a size of 0.4, and is above-median for its seed. But a 1 seed can be a very successful program, like Gonzaga in 2017 with a 42 size, and still be below median for the seed.

Seed	Median size	Seed	Median size	Seed	Median size	Seed	Median size
1	61.7	5	16.1	9	9.3	13	1.0
2	42.0	6	21.3	10	7.3	14	0.7
3	22.6	7	11.2	11	3.2	15	0.5
4	23.3	8	13.0	12	1.7	16	0.3

If a team in a given year had more size than the median for its seed in that year, I labelled it “big” on the grounds that it likely was more familiar to basketball fans and analysts than the “small” teams with below median tournament success. Again I selected a group of 32 big teams, 2 from each seed, and 32 small teams, also 2 from each seed.

For size the direction of the result is surprising. The big, familiar teams everyone knows about win an average of 1.09 games per tournament, versus 0.91 for the small teams, more obscure to basketball fans. 24 of 32 tournaments have been won by schools that were among the top two in size for their seed that year. Apparently, the selection committee does not pay enough attention to prior years' tournament success in setting seeds.

Momentum

For momentum, I looked at the number of wins a team had in its last five regular season games. The table below shows the NCAA tournament winning percentages for each category.⁴ If you look carefully, you'll see that momentum is positive for the top seeds—more wins in the last five regular season games translates to a higher winning percentage in the NCAA tournament. But the pattern is the opposite for the bottom seeds.

If you aggregate the numbers, among the top 8 seeds teams with 0 to 3 wins among their last 5 regular season games win 56% of their tournament games, compared to 67% for top 8 seeds with 4 or 5 wins. But among the bottom 8 seeds, teams with 0 to 3 regular season wins win 37% of their tournament games, versus 22% for bottom 8 seeds with 4 or 5 wins.

I think I know the reason for this. 32 of the NCAA invitations are automatic to winners of 32 conference tournaments. Winners of the 8 strongest conferences (including the Atlantic Coast Conference and Big East) are nearly all teams that would have made the tournament even without the

⁴ If there were fewer than seven games in a category, I omitted the percentage. Only one team ever entered the tournament with five losses in its last five regular season games, a 7 seed that promptly lost its first round game to go to six losses in a row.

conference win. But most of the automatic bids in the other 24 conferences go to teams that only made the NCAA tournament due to the automatic bid.

Seed	Wins in last five regular season games				
	1	2	3	4	5
1			75%	82%	80%
2		59%	70%	72%	74%
3		68%	61%	67%	67%
4		55%	62%	62%	63%
5	44%	58%	47%	55%	55%
6	60%	51%	52%	55%	47%
7	54%	39%	46%	49%	55%
8		37%	47%	38%	38%
9	29%	41%	32%	39%	29%
10	56%	39%	42%	22%	44%
11		54%	35%	40%	33%
12		47%	37%	29%	35%
13			29%	21%	15%
14			21%	15%	15%
15			0%	6%	8%
16			18%	11%	11%

The table below shows the percentage of conference tournament winners among the 16 NCAA tournament seeds. The average is 50%, since half the slots go to the 32 conference winners. But they are not distributed uniformly among the seeds. The best teams in the country usually win their conference tournaments, and are rewarded with 1 seeds in the tournament. Other teams that win good conferences can get top 4 seeds. But teams that are only in the tournament because they won their not-strong conference, are concentrated in the bottom four seeds.

Seed	Conference winners	Seed	Conference winners	Seed	Conference winners	Seed	Conference winners
1	71%	5	23%	9	16%	13	94%
2	43%	6	17%	10	24%	14	97%
3	30%	7	23%	11	46%	15	99%
4	34%	8	16%	12	66%	16	99%

Therefore, my hypothesis is that momentum is good. All other things equal, you prefer a team that's been winning at the end of the regular season. But if a team is only in the tournament due to its momentum, all other things are not equal. Among the bottom 8 seeds, you prefer the 0 to 3 win teams in their last five regular season games, who made the tournament despite a weak finish and no automatic bid.

Value

The three factors above consider only whether a team is likely to win more games than other teams of the same seed. It does not account for the factor that some teams are popular pool selections while

others are dark horses. In equity factors, most of the factors are fundamental measures of the stock, like quality, or technical measures like its price history (momentum factor). But value, measured by price-to-book, is a ratio of market opinion to a fundamental. In NCAA pools, market opinion is measured by pick frequency rather than price.

Most pool participants know pick frequency is important, but they apply it incorrectly. They look for picks with a high ratio of win probability to pick frequency. This is only correct for the largest pools and the most heavily weighted games. For most picks in most pools, it is a poor guide to choices.

To see why, we need to change our focus from picking teams to picking games. We're also going to get a bit more mathy, I apologize.

There are 63 games in a tournament (each game eliminates one team, we need to eliminate 63 of 64 to get a champion). The first round has 32 slots with 4 choices for each. The second round has half as many slots (16) with twice as many choices (4) for each. That pattern continues to the sixth round, the championship game, with 1 slot and 64 choices.

Let's suppose that for each slot we know the probabilities of each potential winning team, and the frequency with which that team will be selected. We can get pretty good estimates of the former from any number of internet sites, and can get some idea of the latter from available statistics and some insight into other pool participants. However even if we knew all those parameters precisely, it's computationally intractable to find the optimal pool; and slight deviations in any parameter leads to totally different choices. So it makes sense to fall back on simple approximations.

Suppose for some slot there are n potential winning teams ($n = 2$ to the power of the round, 2 for round 1 up to 64 for round 6). We call the probability that team i ($i = 0$ to $n-1$) wins p_i and the frequency with which team i is picked f_i . We've labeled the favorite 0, so p_0 is bigger than all other p_i . Of course, both p_i and f_i each sum to 1. We need one other definition, S is the sum of all the $p_i f_i$. That's the expected number of wins this slot will produce for the average pool participant.

The maximum expected value strategy is to choose the favorite in every slot. In fact that will almost always give you an above average point total, 60th to 80th percentile in typical pools, 60th if there are lots of upsets, 80th if there are few. But that will be out of the money in all but the smallest pools. In order to have a chance to win normal-sized pools you have to sacrifice some expected value for increased variance. Clearly you want the choices that give you the maximum ratio of variance gain to expected value loss (actually that's not completely clear, and we'll discuss later when it doesn't work, but it's true for almost all picks in almost all pools).

Suppose you choose team i instead of the favorite, team 0. Your expected number of wins changes by $p_i - p_0$, which is always negative since p_0 is the largest of the p_i s. Your variance relative to the pool average is $p_i(2S+1-p_i-2f_i)-S^2+S$. The largest variance is usually the second favorite. As long as the drop in probability from the favorite to the second favorite is less than twice the drop in frequency—and that is almost always true—the term in parenthesis will be bigger for the second favorite than the favorite. While the initial p that multiplies the term in parentheses will be smaller as well, the product is usually larger for the second favorite than the favorite. The final two S terms are the same for all selections.

It's rarely true that a third or lower favorite will have higher variance than the second favorite, mainly because p drops sharply as you move to less-favored teams. Only occasionally with a third favorite be almost as likely to win as the second, but have a significantly lower frequency of selection. Therefore, people who pick third and lower favorites in the belief that they are increasing the variance of the deviation of their pool total from the mean are mistaken. Picking longshots reduces your expected score, and your expected deviation from average score.

The measure of pick value is therefore not p_i/f_i but the slightly more complicated $p_i + 2(p_0 f_0 - p_i f_i)/(p_0 - p_i) + p_0 - 2S - 1$. The last three terms are the same for all picks, so only the first two matter.

By the way, we use variance rather than standard deviation, because variance adds over picks assuming the slot results are independent.

I mentioned above that maximizing variance given an expected value is not always correct. It assumes a Gaussian distribution of outcomes. That's pretty good for most picks in most pools, each pick value is fairly small compared to the total, the picks are close to (but not exactly) independent and the winner won't be too far out in the tail of the distribution.

But consider a pool so astronomically large that you need to get every pick correct to have any chance of winning. Then p_i measures a pick's contribution to win probability, while f_i measures how many people you'll have to share first prize with, so p_i/f_i is the correct metric. That has little practical relevance, but the same logic applies to picks that you have to win to have much shot at winning a pool.

For example, in most pools scores are weighted such that it's hard to beat someone who gets the right champion if you don't. The larger the pool, and the higher the probability of the champion, the more likely this is true. In very large pools, you may have to get the entire final four right to have much chance, except in very unlikely outcomes than no one predicts.

For "must win" picks, p_i/f_i is the correct metric. For "pretty much must win" picks you can shade the variance metric above toward p_i/f_i . But this has gotten us away from factor investing, toward quant modeling. We're looking for simple additive rules to apply mechanically, not asking about "must win" and "shading."

My "value" picks are for slots rather than teams. In each slot a value pick is on the efficient frontier that maximizes variance from the average pool score for given expected deviation. I set a lower limit of 1 on the ratio, not only must a pick be the most efficient choice available to increase variance, but it must increase variance by at least as much as it reduces expected value.⁵

All 32 tournament winners have been value picks. That is, either they were the favorite or one of the other 63 teams that were picked infrequently enough that it increased variance from the average score by more than it reduced expected values. No underdog with a high pick frequency relative to its probability of winning has ever won the tournament.

However, we can't judge value picks by win percentage, since they are usually favorites or second favorites. We would expect them to win more often than non-value picks. Moreover the point of a value pick is not to win a lot in absolute terms, but to generate wins that move you away from the pool average. That works in both directions, value picks are intended to make you first or last, not above-average.

One way to evaluate value picks would be to apply them to a large selection of pools from previous tournaments. The trouble is that all we would be measuring is the deviation of actual tournament results from probability estimates, and selection frequencies from frequency estimates. If history corresponded exactly to estimates, our value picks would do exactly what was predicted. That's just math.

⁵ Since variance is in different units from expected value, the number 1 is arbitrary. I chose it because it eliminates some very marginal value picks that depend crucially on the accuracy of the probability and frequency estimates.

The other problem is with only 32 actual tournaments we have nowhere near the data necessary to make a reasonable judgment. So on value picks, you have to trust the math (or not).

Betting against beta

My last factor is based on the investment observation that leveraging up low-risk assets (however risk is defined) generally produces better returns than taking the same amount of risk in an unlevered high-risk asset. In bracket pools, this suggests we're better off picking lots of early round minor upsets but the favorite as champion, than fewer early upsets but a less likely champion. Like value, this is a slot-based criterion rather than a team-based one like quality, size and momentum.

For the value picks, we trusted our estimates of pick frequency. Betting against beta is based on other pool participants preferring not to pick too many upsets, instead picking less likely ones. But if we know the frequencies, we don't need to know the reasons. So betting against beta is a factor that can work if our frequency estimates are uncertain.

Another reason betting against beta could work is it gets away from the Gaussian assumptions used in value. It recognizes that picks may not be independent; that someone who makes one upset pick is less likely to make another because people have target numbers of upsets. It's not looking to increase the variance of the deviation of our result from average, but to construct a pool that's different from other people's in the 63-dimensional pick space. The farther we are from other people's brackets, the bigger the region the tournament result can fall into with us the closest.

I chose to measure this with non-favorite picks within 50% of the favorite win probability, and a lower ratio of frequency to probability than the favorite. Unlike value picks, these are not meant to increase variance efficiently, just to differentiate our pools from other people's. We can't test this empirically for the same reason we cannot test value.

Factor correlations

Part of the art of factor construction is selecting a set with low correlations, either positive or negative. Large positive correlations mean the factors usually agree, so they don't add much value to each other. Negative correlations means the factors usually offset, so you seldom make decisions. The five factors work together reasonably well. Quality, momentum and value have moderate positive correlations, size is uncorrelated with anything, and BAB has mild negative correlations.

	Quality	Size	Mom	Value	BAB
Quality	1.00	0.02	0.21	0.24	(0.14)
Size		1.00	(0.01)	0.01	0.04
Mom			1.00	0.20	(0.04)
Value				1.00	(0.12)
BAB					1.00

Still to come

I'm working on a spreadsheet incorporating the NCAA selections for 2019 that were just announced, along with some preliminary data on pool selections. I'll post a spreadsheet to this page (<https://www.eraider.com/nfl-picks>) that you can use to put in information about your pool (size, point system, prize structure) and—if you like—replace my estimates of win probability, pick frequency with your own, or even substitute different measures of quality, momentum and size.